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# Efficacy of "Worm-X" Herbal Formulation against Naturally Infected Gastrointestinal Nematodes in Goats of Banaskantha District, Gujarat

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## ABSTRACT

**Background:** Gastrointestinal nematodosis is a major constraint in economic livestock production. Mostly synthetic anthelmintics are being used to overcome this problem. Besides synthetic anthelmintics, some plants have shown variable anthelmintic properties.

**Methods:** Sixty goats infected with GI nematodes having 1500 EPG were selected and animals were divided into four groups T1 to T4 of 15 animals in each group. Group T1 was given two doses of fenbendazole @ 10 mg/kg BW per os at 15 days interval. In group T2, T3 and T4 herbal formulation "Worm-X" was given @ 0.5, 1.0 and 2.0 ml/kg BW orally respectively, at 15 days interval. Efficacy was assessed on the basis of percentage reduction in EPG count and the FECRT was conducted as per guidelines of WAAVP.

**Result:** When FECR was compared 15 days post treatment, maximum reduction (77.25%) was observed in group T1 followed by group T2 (47.27%), T3 (70.19%) and T4 (72.01%). The reduction in FEC in groupT3 (70.19%) and T4 (72.01%) at 15 days post treatment and at 30 days post treatment were 87.75 and 89.32%, respectively were almost equal. Considering the cost effectiveness and WAAVP recommendations, it will be optimum to administer the drug at the rate of 1.0 ml/kg body weight in two dosages at 15-day interval. However, it is well known that now a day's modern anthelmintic drug efficacy has been reduced in many cases due to the development of anthelmintic resistance. Therefore, complementary or alternative solutions to the conventional chemical treatments have been implied offering novel approaches to the sustainable control of gastrointestinal nematodes in goats.


**Key words:** FECRT, Gastrointestinal nematodes, Goat, Herbal formulation, Worm-X.

## INTRODUCTION

Gastrointestinal nematodes (GIN) infections affect livestock production adversely by reducing body weight gain, meat and milk production (Githiori *et al.,* 2004; Githigia *et al.,* 2005; Gupta *et al.,* 2016). Parasites compete for nutrients with the hosts for their survival resulting in severe clinical signs such as anorexia, anaemia, diarrhoea and oedema. GIN infections are responsible for both direct and indirect major losses, causing decreased productivity, increased costs of control measures and deaths (Sykes, 1994; Torres- Acosta and Hoste, 2008). Up to now the control of GIN was largely based on the repeated use of chemical anthelmintic drugs. For reducing the number of these parasites as well as their adverse effects on production, synthetic anthelmintics are being used but these are not without their side effects. However, it is well known that nowadays their efficacy has been reduced in many cases due to the development of anthelmintic resistance (Gelot *et al*., 2016; Singh *et al.* 2017; Moudgil and Singla 2018; Singh *et al.*2019). The allopathic forms of antihelminthics give adverse effects to the human health as it enters in to the body through animal products, provided no withdrawal period is maintained. Continuous and indiscriminate use of anthelmintic has led to development of anthelmintic resistance as well as drug residues in milk and meat (Singh *et al.* 2002; Pawar *et al.* 2019). Therefore, complementary or alternative solutions to the conventional chemical treatments have been implied offering novel approached to the sustainable control of GIN in small ruminants. This is also supported by an enhanced public concern for more sustainable systems of production, less reliant on the use of chemicals. Besides synthetic anthelmintics, there are some plants or parts of these which have shown known variable anthelmintic properties. In nature also small ruminants have the ability to identify and consume them when suffering from parasitism (Grade *et al.,* 2009). Screening of medicinal plants for their anthelmintic activity nowadays is important to identify herbs with anthelmintic properties and minimize the side effects as well as prevent the problem of anthelmintic resistance. *In vitro* and *in vivo* validation of anthelmintic property of a variety of

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plants has been done by many workers (Arora *et al.*2007; Iqbal *et al.,* 2007; Sunandhadevi *et al.,* 2017). However, before adapting any of these, approaches careful consideration is needed to apply it correctly since small ruminants may demand different application (Papadopoulos, 2008; Torres-Acosta and Hoste, 2008). But scientific validation is needed for anthelmintic properties of these plants. The present study was planned to document the *in vivo* anthelmintic effect of “Worm-X” an herbal formulation on gastrointestinal nematodes of goats.

## MATERIALS AND METHODS

An herbal formulation “Worm-X” manufactured and supplied by ADC Herbal, 240/9, Model Town, Hisar-125 001 (Haryana), India was used to determine the anthelmintic efficacy. Each ml contain crude extract of Kali Jeeri (*Centratherum anthelminticum*) 6.5mg; Indrayan Phal (*Citrullus colocynthis*) 3.5mg; Vidanga (*Embelia ribes*) 6.5mg; Kampilla (*Mallotus philippinensis*) 3.5mg; Ingudi (*Balanites aegyptiaca*) 3.5mg; Palash seeds (*Butea frondosa)* 3.5mg; Neem (*Azadirachta indica*) 3.5mg; Marod Phalli (*Helicteres isora*) 3.5mg; Amaltas (*Casia fistula*) 3.5mg; Kalmegh (*Andrographics paniculata*) 6.5mg Methylparaben-2mg, Propylparaben-3mg and Sodium Benzoate-1.5mg was tested for efficacy against naturally infected gastrointestinal nematode in goats.

### Collection and examination of faecal samples

Faecal samples from 100 adult goats of Banaskantha district of Gujarat were collected from the rectum for parasitological examinations. This area is located in North eastern region of the State were goats are reared by nomadic tribe under ranching system. The district is encompassed by 23.03 to 24.45 Latitude and 71.21 to 73.02 longitudes. The samples were collected in clean labelled sterile vials and taken to the laboratory for microscopic examination by sedimentation methods (Soulsby,1982) Eggs per gram (EPG) of faecal samples were calculated by Modified McMaster method (Coles *et al.*1992).

### Experimental design

Sixty adult goats infected with GI nematodes having EPG more than 1500 were divided into four groups T1 to T4 of 15 animals in each group. Group T1 was given two doses of fenbendazole @10 mg/kg BW per os at 15 days interval. In group T2, T3 and T4 herbal formulation “Worm-X” was given two doses @ 0.5, 1.0 and 2.0 ml/ kg b.wt. Orally, respectively, at 15 days interval. Modified McMaster technique was used for faecal egg counts (Coles *et al.*1992; Zajac and Conboy, 2012). Faecal egg count reduction (FECR) was calculated by using the following formula.

$$\text{FECR (\%)} = \frac{\text{(Pre treatment EPG-Post treatment EPG)}}{\text{Pre treatment EPG}} \times 100$$

In accordance to the recommendations of WAAVP, an anthelmintic is considered as highly effective with FECR percentage more than 98%, effective at 90-98% FECR and moderately effective at 80-89% FECR, whilst FECR percentage less than 80 is not recommended for use. Assessment of therapeutic efficacy of anthelmintics was done on the basis of percentage reduction in faecal egg counts before and 15$^{th}$ and 30$^{th}$ days after treatment.

The experimental data were analyzed by using Hierarchical Design (Snedecor and Cochran, 1994) and the means of various treatment groups were compared using Duncan Multiple Range Test (Duncan, 1955).

## RESULTS AND DISCUSSION

Out of the total animals (100) examined, 80.00% were positive for helminthic infections by concentration methods. Ova of strongyles, *Trichuris* spp. and *Strongyloides* spp. were observed in 65.00, 25.00 and 10.00 % animals, respectively with range 800 to 2400 EPG of nematode infection.

FECR was compared 15 days post treatment, maximum reduction (77.25%) was observed in group T1 followed by groupT4 (72.01%), T3 (70.19%) and T2 (47.27%). The reduction in FEC in group T3 and T4 at 15 days post treatment were almost equal (70.19%) and (72.01%) and at 30 days post treatment were 87.75 and 89.32%, respectively while in group T2 the reduction in FEC at 15 and 30 days post treatment was comparatively less *i.e.* 47.27 and 69.09%, respectively. Results of the study are shown in Table 1. Waghmare *et al.* (2009) reported 100 per cent efficacy of the herbal formulation against GI nematodes in sheep on 13 day post treatment for ten days. The possible explanation for this lower efficacy could be due to difference in animal species as higher dosages are recommended for goats in comparison to sheep. This is due to the fact that the goats metabolize the drugs differently and require a comparatively higher dose (Hutchens and Chappell, 2004). As per WAAVP recommendations, the herbal formulation used in the current study may be classified as not recommended to be used as anthelmintic at a lower dose (0.5 ml/kg) and moderately effective at 1.0 ml and 2.0 ml/kg b wt orally. The herbal formulation used in the present study may be used for effective control of helminths at higher concentrations (1.0 and 2.0 ml/kg body weight) as it was able to reduce more than 85% FEC on 30$^{th}$ day. Considering WAAVP recommendations of such reduction on 15$^{th}$ day; the dose of treatment may be repeated on 15$^{th}$ day after first treatment. The plants used in the herbal formulation have been reported for their anthelmintic activities individually in numerous *in vivo* or *in vitro* studies (Singh *et al.,*1985; Hordegen *et al.,* 2006; Iqbal *et al.,* 2006). Phytochemical studies of the ingredients of the plants have shown that the anthelmintic action of *Butea frondosa* was attributed to the active principle palasonin, phenolic and flavonoids in the seeds, which exerts anthelmintic action through impairment of energy and metabolism by interfering with glucose uptake and by depleting the glycogen stores in the worm (Kumar *et al.,* 1995; Prashant *et al.* 2001; Iqbal *et al.* 2006; Singh *et al.,* 2015). Prashith Kekuda *et al.,* (2009)

**Table 1:** Comparative therapeutic efficacy of fenbendazole and different dose of herbal formulation at different time interval. Figures in parentheses indicate the fecal count reduction per cent.

| Group | Treatment | 0 Days | 15 Days | 30 Days |
|---|---|---|---|---|
| T1 | Fenbendazole @ 10mg/kg PO fortnightly × 2 doses | 2133.33±55.78 | 485.33±20.62 (77.25%) | 98.33±18.17 (95.39%) |
| T2 | Herbal formulation (Worm-X) @ 0.5 mi/kg PO fortnightly × 2 doses | 1613.33±30.65 | 850.68±32.17 (47.27%) | 498.68±25.20 (69.09%) |
| T3 | Herbal formulation (Worm-X) @ 1.0 mi/kg PO fortnightly × 2 doses | 1766.68±48.47 | 526.68±24.82 (70.19%) | 216.33±21.53 (87.75%) |
| T4 | Herbal formulation (Worm-X) @ 2.0 mi/kg PO fortnightly × 2 doses | 1953.33±56.80 | 546.68±43.28 (72.01%) | 208.68±23.64 (89.32%) |

reported that 3 and 5% of aqueous extract of *Embelia ribes* is more potent than the same concentrations of standard drug. Hordegen *et al.,* (2006) reported the ethanolic extracts of seeds of *Azadirachta indica* (Meliaceae), *Caesalpinia crista* (Caesalpiniaceae) and *Vernonia anthelmintica* (Asteraceae), and the ethanolic extracts of the whole plant of *Fumaria parviflora* (Papaveraceae) and of the fruit of *Embelia ribes* (Myrsinaceae) showed an anthelmintic efficacy of up to 93%, relative to pyrantel tartrate. Duvey (2013) reported that that aerial part of Indrayan Phal (*Citrullus colocynthis Syn. Trichosanthes tricuspidata*) ethanolic and aqueous extract of the plant also show significant value for anthelmintic activity in dose dependent manner. *Murali et al.* (2014) reported that *Andrographis paniculata* Nees. (Kalmegh) the medicinal plant, the aqueous extract was found to be more potent, and activities are compared with the drug piperazine citrate as a reference drug. John *et al.* (2009) reported that methanolic extract and its ethyl acetate fraction of *Cassia tora* leaves were evaluated for anthelmintic property using the Indian adult earthworm (*Pheretima posthuma*) as a model. Among the earthworms the ethyl acetate fraction was potent compared with a standard drug, albendazole. The phytochemical analysis of both extracts showed the presence of phenolics like flavonoids and tannins as well as anthraquinones, which may be the active principle for anthelmintic property. Chothani and Vaghasiya (2011) reported that *Balanites aegyptiaca* known as ‘Ingudi,’ traditionally used in treatment of various ailments including intestinal worm infection (Nkunya *et al.*, 1990). Some components of the seed of *Centratheram anthelminticum,* such as vernodlin, vernodalol and vernolic acid, have been isolated and identified (Asaka *et al.,* 1977; Lambertini *et al.,* 2004) and are known to have bitter taste and these bitter principles may be responsible for the anthelmintic activity. Triterpenol derivatives of *Azadirachia indica* (Neem) like azadirachtin have shown anthelmintic properties (Verma *et al.,* 1995). Arora *et al.* (2007) found 89.95% reduction in faecal egg count after using hot methanolic extract of neem leaves (*Azadirachia indica*) @150mg/kg b.wt. In conclusion, this seems to be this herbal formulation “Worm-X” in goats as effective anthelmintic for control of gastrointestinal nematode infection. The efficacy of herbal formulation used in present study was optimum at either of the higher dosage (1.0 and 2.0 ml/kg body weight). Considering the cost effectiveness and WAAVP recommendations, it will be optimum to administer the drug at the rate of 1.0 ml/kg body weight in two dosages at 15-day interval.

## CONCLUSION

Our study summaries the plant derived products as efficient and can be prescribed against gastrointestinal parasites in goats to counter the threat of looming resistance development. Antihelminthic effect of the extract may be attributed to the phytochemical constituents such as sterols and terpenes, polyphenols, flavonoids, tannins saponins, and alkaloids. These compounds would be having strong activity due to the phytochemicals and could explain the antihelminthic activity of the plant. However, the more detailed phytochemical analysis is required to isolate and characterize each active compound, which is responsible for the antihelminthic activity and exact mechanisms of action of this activity.

The results of the present study revealed the potential use of the plant derivatives against the worms. The extract was found to be even more effective in higher concentrations than the standard drug. The results are in justification with the traditional use of the plant for exclusion of worms. There is further need for more phytochemical screening and their antihelminthic effect should be determined for development of better formulation to control the gastrointestinal parasites in livestock.

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